Mastering the Harmonic-Wise Pipeline: From Complex Signals to Compact Descriptors

In the high-stakes environment of industrial signal processing, understanding Transmission Error (TE) is critical for system health and compensation. However, how we choose to represent that signal determines the success of our machine learning models. This guide details the transition from raw, point-by-point data to a structured harmonic-wise pipeline—a shift that prioritizes the mechanical "recipe" of a gearbox over the noise of individual data samples.

1. The Core Philosophy: Why Summarize a Curve?

Industrial periodic signals are governed by the physics of the hardware. Traditional methods often attempt to predict the TE value at every specific angular position. This approach is frequently inefficient and susceptible to noise. The harmonic-wise pipeline represents a fundamental shift: we move from predicting points on a curve to predicting a compact **summary** of the entire curve.**The "So What?":** For industrial periodic signals, the underlying periodic structure matters significantly more than individual, noise-heavy data points. By summarizing a curve into its constituent harmonics, we capture the "mechanical DNA" of the hardware, providing the model with a structured representation that is inherently more resilient to signal fluctuations.

2. Paradigm Shift: Direct-TE vs. Harmonic-Wise Modeling

To appreciate the strategic value of this pipeline, we must contrast it with the standard Direct-TE approach (TE-CVP) versus the paper-faithful Harmonic-Wise approach (RCIM-MBR).

Feature	Direct-TE Approach (TE-CVP)	Harmonic-Wise Approach (RCIM-MBR)
Input Data	Angular position + Operating conditions	**Operating conditions only** (Speed, Torque, etc.)
Target Variable	A single TE value at a specific angle	**A compact set of harmonic coefficients**
Model's Primary Task	Predicting a point on a curve	**Predicting the "recipe" of the entire curve**

This paradigm shift means we no longer treat the angle as an input; instead, the entire angular cycle is the object we are decomposing and describing.

3. The Decomposition Process: Turning Signals into Ingredients

The process of "summarizing" the curve is technically a **truncated harmonic representation**. We treat the TE signal as a summation of specific periodic components that align with the practical periodic structure of the mechanical system.

The Mathematical Representation

Conceptually, the curve is decomposed using a sum of sines and cosines:

```
TE(theta) ≈ a0 + sum_k [c_k * cos(k * theta) + s_k * sin(k *
theta)]
```

- **a_0** : The DC offset or baseline of the curve.
- **c_k** **and s_k** : The cosine and sine coefficients for each harmonic order k .

The Selected Harmonics

We do not use every possible frequency. Instead, we select a specific set of harmonics that represent the mechanical frequency response of the gear system as defined in the paper-oriented comparison scope:

- **Fundamental & Low Orders:** 0, 1, 3
- **Mid-Range Orders:** 39, 40, 78, 81
- **High Orders:** 156, 162, 240These specific orders are chosen because they capture the actual periodic signatures inherent in the hardware's operation.

4. Coefficients, Amplitude, and Phase: The Language of Harmonics

While the internal math relies on coefficient pairs (c_k, s_k), physical analysis often focuses on **Amplitude** (A_k) and **Phase** (ϕ_k).

- **Internal Math vs. Physical Reality:**
- **Repository Internal Math:** The pipeline predicts coefficient pairs (c_k, s_k). This is a strategic choice for **stable regression** . Predicting Phase (ϕ_k) directly is notoriously difficult for standard regressors due to "non-linear phase wrapping"—the discontinuities that occur when a phase jumps from 2π back to 0 . Sine/Cosine pairs avoid this issue entirely, providing a continuous target for the model.
- **Paper Physical Quantities:** These coefficients map directly to the physical descriptors used in the reference paper via these formulas:
- **Amplitude:** $A_k = \sqrt{c_k^2 + s_k^2}$
- **Phase:** $\phi_k = \operatorname{atan2}(-s_k, c_k)$By predicting the coefficients, we maintain signal integrity while remaining fully compatible with the physical interpretations of amplitude and phase.

5. Building the Supervised Dataset: The "Key Conceptual Shift"

The most significant architectural change occurs in Stage 4: the restructuring of the supervised learning task. We move away from high-density, point-wise datasets toward a tabular "harmonic summary" format.**The Input (Operating Conditions):** The model focuses exclusively on the environmental and operational variables:

- **Speed**
- **Torque**
- **Temperature**
- **DirectionThe Target (Harmonic Summary):** Instead of a single scalar TE value, the target is the full set of harmonic ingredients (the c_k and s_k coefficients) that describe the curve produced under those specific conditions. This transforms a complex signal-processing problem into a clean, multi-target regression task.

6. Reconstruction and Evaluation: Closing the Loop

Once the model (currently a **HistGradientBoosting** regressor) predicts the coefficients, we must translate these abstract numbers back into physical reality through **TE Reconstruction** . This serves as the bridge, rebuilding the full TE curve over the entire angular cycle so it can be compared against the ground truth.**Evaluation Metrics:** To validate the reconstruction, we use three primary metrics: **MAE** , **RMSE** , and **Mean Percentage Error** (the primary paper-aligned benchmark).**Current Status: The current baseline achieves a Mean Percentage Error of ~9.4%. While the ultimate industrial**

goal (Target A) is $\le 4.7\%$, our "Oracle" tests—using perfect harmonic data—achieve errors far below this threshold. This is a vital insight: the selected harmonic set is expressive and sufficient; the current bottleneck is the predictor's accuracy, not the signal-processing methodology.

7. Summary: The Path Forward

The harmonic-wise pipeline is more than just a different way to organize data; it is a strategic route toward interpretable, structured industrial AI.

1. **Compact & Robust Representation:** By predicting curve "descriptors" rather than individual points, the learning problem becomes more manageable and less sensitive to sample-rate noise.
2. **Structured Physical Modeling:** The use of selected harmonics ensures the model remains grounded in the mechanical realities of the hardware.
3. **Path to Paper-Faithful Benchmarking:** This framework provides a clear, reproducible roadmap to hit the required industrial thresholds, allowing for a genuine comparison with state-of-the-art research.**What This Is Not:** This pipeline is currently an **offline-only** research branch. It is a critical benchmarking and validation tool used to prove the feasibility of the harmonic approach, but it is not yet a real-time online compensation system or a hardware deployment package.